1 导数

$$f'(x_0)=\lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}=\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$$

物理含义：速度，变化率 $f'(x_0)=\lim_{\Delta x\to 0}\frac{\Delta f}{\Delta x}$

几何含义 graph(f)上 $P(x_0,f(x_0))$ 处切线的斜率

切线方程 $y-f(x_0)=f'(x_0)(x-x_0)$

例 $\lim_{n\to\infty} n\left(f(x+\frac{1}{n})-f(x)\right)=\lim_{n\to\infty}\frac{f(x+\frac{1}{n})-f(x)}{\frac{1}{n}}\overset{\text{Heine}}{=\!=}\lim_{t\to 0}\frac{f(x+t)-f(x)}{t}=f'(x)$

$\mathbb{Z}_+ \xrightarrow{\frac{1}{n}} \mathbb{R}\setminus 0 \xrightarrow{g} \mathbb{R}$

$g(t)=\frac{f(x+t)-f(x)}{t}$

$g(\frac{1}{n})=\frac{f(x+\frac{1}{n})-f(x)}{1/n}$

用于 $f(x)=\ln F(x)$ 得

$$\lim_{n\to\infty} n\ln\frac{F(x+\frac{1}{n})}{F(x)}=f'(x)=\frac{1}{F(x)}F'(x)$$

（Chain Rule）

$$\lim_{n\to\infty}\ln\left(\frac{F(x+\frac{1}{n})}{F(x)}\right)^n$$

两边 $e^{\square}$ $\Rightarrow$

$$\lim_{n\to\infty} e^{\ln\left(\frac{F(x+\frac{1}{n})}{F(x)}\right)^n}=e^{F'(x)/F(x)}$$

$$\lim_{n\to\infty}\left(\frac{F(x+\frac{1}{n})}{F(x)}\right)^n$$

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2 计算

- 依定义
- 四则运算
- 复合函数求导 $\leftrightarrow$ Chain Rule 链式法则
- 反函数求导

(四则运算) Thm 设 f, g 在 x_0 处可导, 则

① $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$

② (Leibniz 法则) $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$

③ 当 $g(x_0) \neq 0$ 有 $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$

证 ②
$$(fg)'(x_0) = \lim_{x \to x_0} \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0}$$

$$= f'(x_0)g(x_0) + f(x_0)g'(x_0)$$

(由 g 可导 $\Rightarrow g$ 连续)

③
$$\left(\frac{f}{g}\right)'(x_0) = \lim_{x \to x_0} \frac{f(x)g(x_0) - f(x_0)g(x_0) + f(x_0)g(x_0) - f(x_0)g(x)}{(x - x_0)\, g(x)g(x_0)}$$

$$= \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2} \quad \#$$

推论 设 f, g 处处可导, 则

① 求导是线性算子 $(\lambda f + \mu g)' = \lambda f' + \mu g'$ $\forall \lambda, \mu \in \mathbb{R}$

② 求导满足 Leibniz rule $(fg)' = f'g + fg'$

(称满足此3个性质的算子为导子)

③ $C' = 0$

④ $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ → $\frac{(\text{分子})'(\text{分母}) - (\text{分子})(\text{分母})'}{(\text{分母})^2}$

推论 (多次 Leibniz)

$$(f_1 f_2 \cdots f_k)' = f_1' f_2 \cdots f_k + f_1 f_2' \cdots f_k + \cdots + f_1 \cdots f_{k-1} f_k'$$

证. $(f_1 \cdots f_k)' = (f_1 (f_2 \cdots f_k))' = f_1' (f_2 \cdots f_k) + f_1 (f_2 \cdots f_k)'$

$=$ 用归纳假设即可 #

例 • $C' = 0$ (常值函数)

• $x' = 1$

• $(x^n)' = \lim\limits_{h \to 0} \dfrac{(x+h)^n - x^n}{h} = C_n^1 x^{n-1} = n x^{n-1}$

$(n \in \mathbb{Z}_+)$ (下放指数为因子, 且指数降1)

• polynomial

$$\left(\sum_{i=0}^{n} a_i x^i \right)' = \sum_{i=0}^{n} a_i i x^{i-1} = \sum_{i=1}^{n} a_i i x^{i-1}$$

• 若 $f(x) = a_n (x - x_1) \cdots (x - x_n)$

$$f'(x) \xlongequal{\text{多次 Leibniz}} a_n \Big((x-x_2) \cdots (x-x_n) + (x-x_1)(x-x_3) \cdots (x-x_n) + \cdots + (x-x_1) \cdots (x-x_{n-1}) \Big)$$

$$= f(x) \sum_{i=1}^{n} \frac{1}{x - x_i}$$

多复函数 $\oint_C \dfrac{f'(x)}{f(x)} = \oint_C \sum\limits_{i=1}^{n} \dfrac{1}{x - x_i}$ → 探测出 f 的根的信息

• 定义 称 x_0 是多项式 $f(x)$ 的 m 重根

$\Leftrightarrow$ $f(x) = (x - x_0)^m g(x)$ 且 $g(x_0) \neq 0$

事实 若 x_0 是 f 的 m (≥ 1) 重根, 则 x_0 是 f' 的 $m-1$ 重根

因为 $f(x)' = \big((x - x_0)^m g(x) \big)'$

$= m (x - x_0)^{m-1} g(x) + (x - x_0)^m g'(x)$

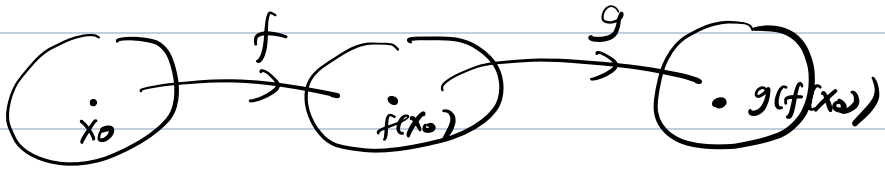
$$= (x-x_0)^{m-1}\big(m g(x) + (x-x_0) g'(x)\big)$$

$$= (x-x_0)^{m-1} \; h(x)$$

$\Rightarrow h(x_0) = m g(x_0) + 0 = m g(x_0) \neq 0$

$\Rightarrow x_0$ 是 f' 的 $m-1$ 重根

复合函数求导



$$(g\circ f)'(x_0) = \lim_{x\to x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} \quad \leftrightarrow \Delta g$$

当 $f(x)\neq f(x_0)$ 此式无意义 ←

$$\overset{?}{=} \lim_{x\to x_0} \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0}$$

$$= \lim(\text{前一个因子}) \cdot \lim(\text{后因子})$$

$$\overset{\text{换元 } y=f(x)}{=\!=} \left(\lim_{y\to f(x_0)} \frac{g(y) - g(f(x_0))}{y - f(x_0)}\right) \cdot f'(x_0)$$

手跳记 修正图/面？

f 可导 $\Rightarrow$ f 连续

当 $x\to x_0$, $y = f(x) \to f(x_0)$

$$\overset{\text{(定义)}}{=\!=} g'(f(x_0))\, f'(x_0)$$

Need 手把改写（手作不等式估计）→（练习）

马上用微分观点重记

Thm. (Chain Rule) 设 f 在 x_0 处可导

g 在 $f(x_0)$ 处可导

则 $g\circ f$ 在 x_0 处可导且

$$(g\circ f)'(x_0) = g'(f(x_0)) f'(x_0)$$

#

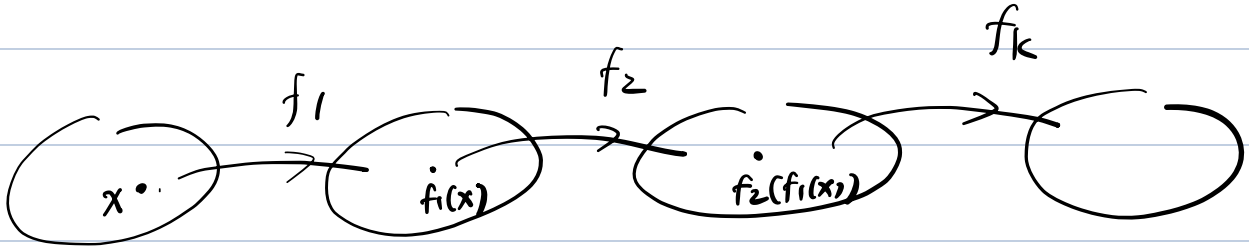
推论 设 f, g 处处可导，则

$$(g\circ f)'(x) = g'(f(x))\, f'(x).$$

左边写成 $\big(g(f(x))\big)'$，或 $\frac{d}{dx}\big(g(f(x))\big)$ 也可接受

推论（多重复合）

$$\big(f_k\circ\cdots\circ f_2\circ f_1\big)'(x) = f_k'\big(f_{k-1}\circ\cdots\circ f_1(x)\big)\cdots\cdot f_2'(f_1(x))\, f_1'(x)$$



记 对 k 归纳

$$\text{LHS} = \big(f_k\circ(f_{k-1}\circ\cdots\circ f_1)\big)'(x)$$
$$= f_k'\big(f_{k-1}\circ\cdots\circ f_1(x)\big)\cdot\big(f_{k-1}\circ\cdots\circ f_1\big)'(x)$$
$$= \text{用归纳假设} \qquad \#$$

例 • $(\sin x)' = \lim\limits_{h\to 0}\dfrac{\sin(x+h)-\sin x}{h} = \lim\limits_{h\to 0}\dfrac{\cos(x+\frac{h}{2})\sin\frac{h}{2}}{\frac{h}{2}}$

$$= \cos x$$

• $(\cos x)' = \lim\limits_{h\to 0}\dfrac{\cos(x+h)-\cos x}{h} = \lim\limits_{h\to 0}\dfrac{-2\sin(x+\frac{h}{2})\sin\frac{h}{2}}{h}$

$$= -\sin x$$

• $(\tan x)' = \left(\dfrac{\sin x}{\cos x}\right)' = \dfrac{\cos x\cos x - \sin x(-\sin x)}{\cos^2 x} = \dfrac{1}{\cos^2 x}$

• $(\cot x)' = \left(\dfrac{\cos x}{\sin x}\right)' = \dfrac{(-\sin x)\sin x - \cos x\cdot\cos x}{\sin^2 x} = \dfrac{-1}{\sin^2 x}$

- $(e^x)' = \lim_{h\to 0} \frac{e^{x+h}-e^x}{h} = e^x \lim_{h\to 0} \frac{e^h-1}{h}$

$\overline{e^h-1=y}$ $e^x \lim_{y\to 0} \frac{y}{\ln(1+y)} = e^x \lim_{y\to 0} \frac{1}{\ln(1+y)^{1/y}}$

$= e^x \frac{1}{\ln e} = e^x$.

(和 $\frac{\partial}{\partial y}(e^x) = 0$)

- $(\ln x)' = \lim_{h\to 0} \frac{\ln(x+h)-\ln x}{h} = \lim_{h\to 0} \frac{1}{x} \ln\left((1+\frac{h}{x})^{\frac{x}{h}}\right)$

$= \frac{1}{x} \ln e = \frac{1}{x}$

例1 $\left(\sin(f(x))\right)' = \cos(f(x)) \cdot f'(x)$

$\left(e^{f(x)}\right)' = e^{f(x)} \cdot f'(x)$

$\left(\ln(f(x))\right)' = \frac{1}{f(x)} f'(x)$

$\left(\ln|x|\right)' = \frac{1}{|x|}\left(|x|\right)'$

$= \frac{1}{|x|} \cdot \begin{cases} 1 & x>0 \\ -1 & x<0 \end{cases}$

$= \frac{1}{x}$ $(\forall x \neq 0)$

$\boxed{\left(\ln|x|\right)' = \frac{1}{x} \quad \forall x \neq 0}$

進而 $\left(\ln|f(x)|\right)' = \frac{1}{f(x)} f'(x)$

- $(x^\alpha)' = \left(e^{\alpha \ln x}\right)' = e^{\alpha \ln x} (\alpha \ln x)'$

$\forall \alpha \in \mathbb{R}$ $= e^{\alpha \ln x} \alpha \frac{1}{x} = \alpha x^{\alpha-1}$

$$\boxed{(x^\alpha)' = \alpha x^{\alpha-1} \quad \forall \alpha}$$

- $$(u(x)^{v(x)})' = (e^{v(x)\ln u(x)})'$$
$$= e^{v\ln u}(v\ln u)'$$
$$\underset{\text{Leibnit}}{=} e^{v\ln u}(v'\ln u + v\cdot\frac{1}{u}u')$$

$$(x^{x^x})' = (e^{x^x\ln x})' = e^{x^x\ln x}(x^x\ln x)'$$
$$= x^{x^x}\left((e^{x\ln x})'\ln x + x^x\frac{1}{x}\right)$$
$$= x^{x^x}\left(e^{x\ln x}(x\ln x)'\ln x + x^{x-1}\right)$$
$$= x^{x^x}\left(e^{x\ln x}(\ln x+1)\ln x + x^{x-1}\right)$$
$$= x^{x^x}\left(x^x(\ln x+1)\ln x + x^{x-1}\right)$$

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3 微分

一元函数：可微 = 可导

多元函数：可微 $\Rightarrow$ 可导 且可微更基本概念

f在x_0处可导 $\Longleftrightarrow$ $\lim\limits_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$ $\exists$

$\Longleftrightarrow$ $\exists A\in\mathbb{R}$. $\lim\frac{f(x_0+h)-f(x_0)}{h} = A$

$\Longleftrightarrow$ $\exists A\in\mathbb{R}$ $\lim\frac{f(x_0+h)-f(x_0)-Ah}{h} = 0$

$\Longleftrightarrow$ 余记为$\alpha(h)$

$\exists A\in\mathbb{R}$ 使 $f(x_0+h) = f(x_0)+Ah+\alpha(h)$

且 $\lim\limits_{h\to 0}\frac{\alpha(h)}{h} = 0$

$$f(x_0+h) = f(x_0) + Ah + \alpha(h)$$

↓ 0次　↓ 线性项　↓ 误差项

之所以称 α 为误差 因为 $\lim \frac{\alpha(h)}{h} = 0$，比一次项显著小

$\alpha(h) = o(h)$

即在 x_0 附近 f 可近似为一次(线性)函数

⟶ 可通过线性化来研究 f 在局部的行为

(f 近似 2 次函数，二次型

n 阶 Taylor 式．张量 描述 n 次齐次多项式)

定义 称 f 在 x_0 处可微，若 ∃ 线性映射 $L: \mathbb{R} \to \mathbb{R}$

↕ Linear

使 $f(x_0+h) = f(x_0) + L(h) + \alpha(h)$

且 $\lim_{h\to 0} \frac{\alpha(h)}{h} = 0$ (即 $\alpha(h) = o(h)$)

进一步地满足此条件的唯一的线性映射 L 为 f 在 x_0 处的微分，记为 $df_{x_0}: \mathbb{R} \to \mathbb{R}$

(微分 df_{x_0} 是一个线性映射.

微分是 f 的线性近似

(不提 $f(x_0)$))

Prop 对一元的 f 有

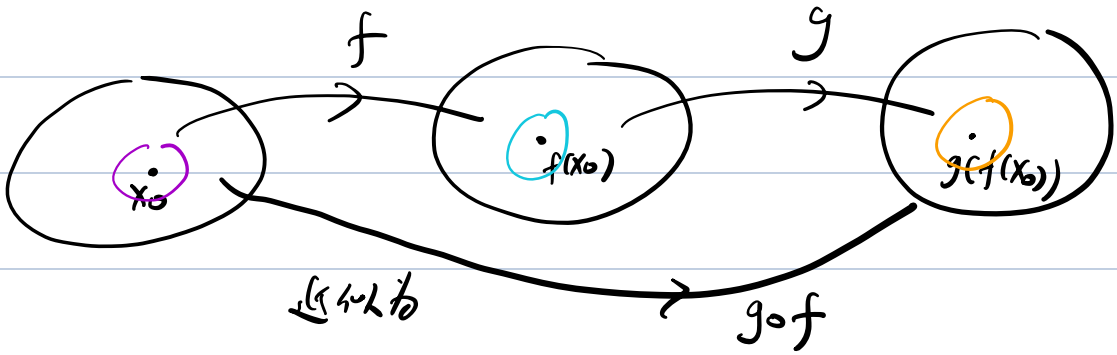
f 可微 $\Leftrightarrow$ f 可导

且 f 在 x_0 处 的微分为 ~~$df_{x_0}(h) = L(h) = Ah = f'(x_0)h$~~

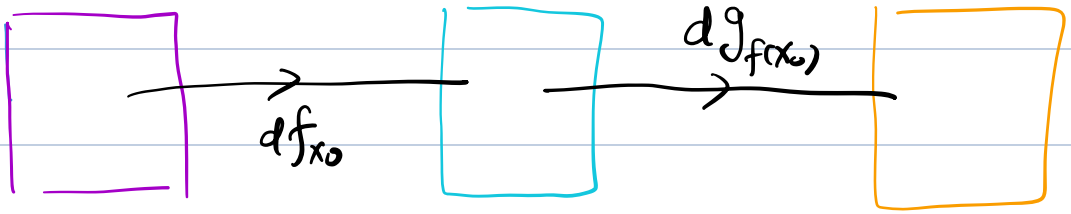
$df_{x_0}(h) = f'(x_0)h \quad \forall h \in \mathbb{R}$

(微分的系数矩阵为 $(f'(x_0))_{1\times 1}$) #

理解 Chain Rule ?



显微镜



想 $g\circ f \simeq dg_{f(x_0)} \circ df_{x_0}$ (此条件是线性的)

$\Rightarrow d(g\circ f)_{x_0} = dg_{f(x_0)} \circ df_{x_0}$

Thm (复合的微分等微分的复合)

设 f 在 x_0 处可微, g 在 $f(x_0)$ 处可微

则 $g\circ f$ 在 x_0 处可微 且

$d(g\circ f)_{x_0} = dg_{f(x_0)} \circ df_{x_0}$ (*)

与 Chain Rule 何关？ ⊛是线性映射之间"="式

作用在 $h \in \mathbb{R}$ 上

$$d(g\circ f)_{x_0}(h) \overset{\circledast}{=} dg_{f(x_0)} \circ df_{x_0}(h) \qquad \forall h$$
$$= g'(f(x_0))\left(f'(x_0)h\right)$$

$$\| \quad (g\circ f)'(x_0)h$$

$$\Rightarrow \quad (g\circ f)'(x_0) = g'(f(x_0))\cdot f'(x_0)$$

(Chain Rule) #

Thm 证明 设 $df_{x_0}(h) = Ah \quad \forall h$

$dg_{f(x_0)}(v) = Bv \quad \forall v$

由于微分定义 $f(x_0+h) = f(x_0) + Ah + \alpha(h)$ 且 $\lim_{h\to 0}\frac{\alpha(h)}{h} = 0$

$g(f(x_0)+v) = g(f(x_0)) + Bv + \beta(v)$ 且 $\lim_{v\to 0}\frac{\beta(v)}{v} = 0$ ← $\beta(0)=0$

$$\Rightarrow \quad g\circ f(x_0+h) = g\left(f(x_0) + \underbrace{Ah+\alpha(h)}_{v}\right)$$

$$\overset{\circledast}{=} g(f(x_0)) + B(Ah+\alpha(h)) + \beta(Ah+\alpha(h))$$

$$= g(f(x_0)) + BAh + B\alpha(h) + \beta(Ah+\alpha(h))$$

$g(f(x_0))$：0次　BAh：线性　$B\alpha(h)+\beta(Ah+\alpha(h))$：误差？

只需证 $\lim_{h\to 0}\frac{B\alpha(h)+\beta(Ah+\alpha(h))}{h} = 0$?

$$\Leftrightarrow \lim \frac{\beta(Ah+\alpha(h))}{h} = 0 \quad \text{(★)}$$

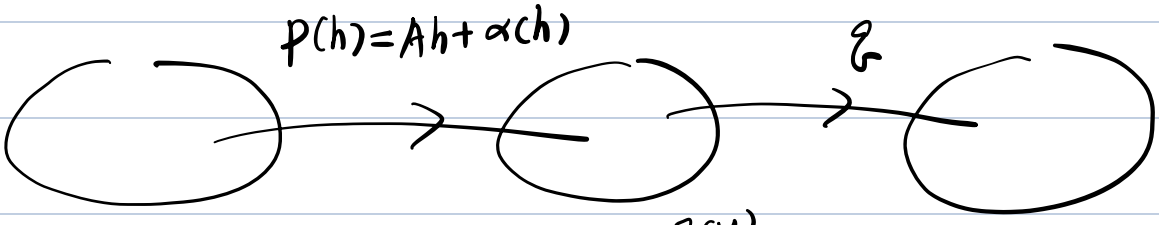
(若如此引⊛说 $g\circ f$ 在 x_0 处可微

且 微分为 $d(g\circ f)_{x_0}(h) = BAh$

$= d\varphi_{f(x_0)}(df_{x_0}(h))$ //

$$\frac{\beta(Ah+\alpha(h))}{h} \overset{\text{rope}}{=} \frac{\beta(Ah+\alpha(h))}{Ah+\alpha(h)} \cdot \frac{Ah+\alpha(h)}{h}$$

(当 $Ah+\alpha(h)=0$ 非法)



$$g(v)=\begin{cases}\frac{\beta(v)}{v}, & v\neq 0\\ \lim_{v\to 0}\frac{\beta(v)}{v}=0, & v=0\end{cases}$$

则 g 在 0 处连续 $\Rightarrow$ 满足假设 (II)

$$\Rightarrow \lim_{h\to 0} g(p(h)) = \lim_{v\to 0} g(v) = 0$$

$$\Rightarrow \lim \; g(p(h)) \; \frac{Ah+\alpha(h)}{h} = 0\cdot A = 0$$

(因此)

$g(p(h)) \to 0$, $\frac{Ah+\alpha(h)}{h} \to A$

易 check

$$g(p(h))\frac{p(h)}{h} = \begin{cases}\frac{\beta(p(h))}{h} & p(h)\neq 0\\ 0=\frac{\beta(p(h))}{h} & p(h)=0\end{cases}$$

(因为 $\beta(0)=0$)

$$= \frac{\beta(Ah+\alpha(h))}{h} \quad \forall h$$

因此 $\lim_{h\to 0}\frac{\beta(Ah+\alpha(h))}{h} = \lim_{h\to 0} g(p(h))\frac{p(h)}{h} = 0\cdot A = 0$

#